Cross-Sectional Machine Learning for Long-Short Equity Selection

An automated multi-asset panel forecasting & portfolio-construction pipeline

---

1. Objective, Target & Data

Objective. Build an automated, causally-correct machine-learning workflow that forecasts short-horizon *cross-sectional* relative performance across a panel of equities and translates those forecasts into a systematic **dollar-neutral long-short** portfolio, evaluated with a full statistical-significance battery.

Universe (20 names). A balanced panel of mid- and small-cap U.S. equities drawn from the provided lists — deliberately **avoiding large-caps, ETFs, and the Magnificent 7**, whose passive-flow-driven prices carry little technical signal:

- **Mid-cap:** FLEX, TWLO, FTI, CW, NVT, STRL, XPO, EWBC, RGA, EVR
- **Small-cap:** BLKB, CNXC, CACC, LAZ, EBC, WSBC, WRBY, LQDA, RSI, USLM

Target (single quantity, panel-wide, regression). For each stock i and day t, predict the **1-day-forward market-relative log return**:

$$y[i,t] = r[i,\, t \to t+1] - \text{mean}_j\, r[j,\, t \to t+1] \quad \text{where} \quad r = \log(C_{t+1} / C_t)$$

Regression (not a median-split classification) is chosen so that **MAE** and **directional accuracy by quantile** are well-defined; cross-sectional demeaning makes the target market-neutral and connects directly to the CAPM-alpha evaluation.

Frequency & window. Daily. Data pulled `2022-01 → 2026-06` (~4.5 years); `2022-01` is the earliest date at which *all 20 names* trade, giving a fully balanced panel. The final **12 months are held out as an untouched test set**; the first ~90 rows are dropped as feature burn-in, and a ~5-day embargo separates train from test.

Source. WRDS — CRSP (OHLCV + bid/ask), OptionMetrics (put-call), short-interest, plus a market index and VIX. Restructured into a hierarchical `MultiIndex(Date, Ticker)` frame for vectorized cross-sectional transforms.

2. Feature Engineering & Scaling

Indicators (18 + a calendar block; >= 7 exogenous). Two buckets — the requirement to include indicators *not* based on the target's own OHLCV bar is met by the second:

Bucket	Indicators
Target-OHLCV (TA-Lib)	RSI(14), MACD-hist, PPO, ROC(10), ATR/Close, Bollinger %position, ADX(14), relative volume, CMMA_10, realized-vol(20d), momentum 5d & 21d
Exogenous (not the target's bar)	sector-relative 21d return, 60d market **beta**, 60d **idiosyncratic vol**, **put-call ratio**, **short-interest** days-to-cover, **VIX** 5d change, **bid-ask** quoted spread, **calendar set** (day-of-week one-hot + month sin/cos)

Scaling (two causal stages).

1. *Element-wise transforms* — independent-row scaling for TA-Lib indicators (e.g. `RSI → (RSI-50)/50`, `MACD → asinh(MACD/Close)`, `PPO → /100`, `ATR → log1p(ATR/Close)`), log-first for positive multiplicative series, `asinh` for signed series.
2. *Per-day cross-sectional z-scoring* — in a panel this is the natural scaling **and** is inherently look-ahead-free (same-day only).

A `StandardScaler` additionally sits **inside** the CV pipeline for the scale-sensitive models. The calendar set is left unscaled and selected/dropped as a unit.

Feature selection — discard the worst 4. Rank every feature by **univariate Profit Factor** (Alpha >= 0.30 method, re-implemented in Python since `VarScreen.exe` is Windows-only), drop the bottom 4, and cross-check against training-set mutual information. `SelectKBest` remains a tunable step within the scaled-model pipelines. PACF informs the choice of autoregressive return lags.

3. Modelling

Roster (5 models, >= 1 requiring scaling). LightGBM, RandomForest, ElasticNet, SVR, **Keras MLP** (the last three scaled).

Selection & tuning. A **manual grid-search cross-validation loop** compares all five under a **date-grouped `TimeSeriesSplit`** (each day's full cross-section kept on one side of every fold), scored by a `make_scorer` on **cross-sectional Rank IC** (mean daily Spearman). The winner is then regularized/fine-tuned with `GridSearchCV` / `RandomizedSearchCV`.

Neural-network specifics. The MLP is trained with **MSE loss + L2 weight penalty**, **sample weights proportional to |target|** to emphasize large movers, and dropout + early-stopping for regularization.

Causal backbone (the make-or-break). All features and targets are built per-ticker via `groupby`; the target is a **per-ticker `shift(-1)` forward return** (never a global shift, which would bleed across tickers); burn-in dropped; train/test embargoed; CV grouped by date. This gets its own validated section in the notebook.

4. Evaluation & Strategy

Strategy (plain English). Each day, rank the 20 names by predicted relative return; **go long the top quintile (4 names) and short the bottom quintile (4 names)**, equal-weighted and dollar-neutral, rebalanced daily with a ~5-10 bps round-trip cost. This trades the **extreme quantiles** the guidelines call for.

Stage	Reported
Validation (out-of-fold)	MAE, Profit Factor, Spearman IC, directional accuracy by quantile vs majority-direction baseline
Test (held-out 12 mo)	the four above **+ SHAP** feature-importance on the winning model
Equity curve (test)	long-short vs equal-weight buy-and-hold
Diagnostics (test)	**White's Reality Check** p (log-return-adapted), **Monte-Carlo permutation** p (Profit-Factor statistic), Sharpe, Profit Factor, CAGR, CAPM alpha/beta, annualized volatility

5. Deliverables, Timeline & Limitations

Deliverables. A single Jupyter notebook, all WRDS data bundled as CSV, and a 30-50 slide deck with extensive speaker notes.
Stack: scikit-learn, LightGBM, Keras/TensorFlow (no PyTorch), Python <= 3.11.

Timeline (~20 days).

Days	Work
1-6	Data pull + causal feature/target construction (front-loaded — highest-risk step)
7-12	Feature screen, model grid-search, hyperparameter tuning
13-16	SHAP, trading, equity curve, significance tests
17-20	Slide deck with speaker notes

Limitations (stated openly). A static current-universe backtest carries mild survivorship bias; small-cap options/short-interest coverage can be sparse (fallback: drop the feature for over-sparse names); 20-name breadth makes daily long-short PnL noisy. Per the brief, **the equity curve need not be profitable — correctness of the pipeline logic is the objective.**